Biochemistry Cume April 3, 2015
Polypeptide/Protein Structure, Analysis

1) (40 pts.) Examine the information below. Indicate whether the following peptides would be cleaved by the indicated enzymatic or chemical treatment. If cleavage would take place, briefly describe where the cleavage would occur and give the products of the cleavage reaction.

Peptide	Treatment
Phe-Arg-Pro-Tyr-Met-Ile	Chymotrypsin
Trp-Met-Leu	Trypsin
Gly-Met-Pro-Cys-His-Met-Arg	CnBr
Pro-Arg-Lys-Gly-Ala-Ala-Lys	Trypsin

2) (15 pts) Explain one or more ways to identify and separate any of the products produced above. Explain the basis for the separation.

3) (15 pts) Histones are an important class of proteins that bind to DNA. Histones have a high percentage of Arg and/or Lys in their sequences (15-35%). Does this finding make it easy to determine histone sequences or not? Briefly explain your answer.

4) (30 pts.) Examine the following amino acid residues below. Which residues would you expect to find on the inside, and which would you expect to find on the outside, of a typical globular protein molecule in a solution with a pH = 7? For each, briefly explain your rationale.

Glu	**Arg**
Val	**Phe**
Ile	**Gly**
Asn	**Lys**
Ser	**Thr**